RESEARCH UPDATE · 1 NOV 2026

The Tests Passed.
The Benchmark Said No.

What if the grader of an AI security fix is wrong?

TESTS PASS → FINAL VERDICT FAIL

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THE IDEA

Imagine a teacher with a checklist

THE CHECKLIST

✓ **All listed checks pass**

The student's work satisfies what the checklist actually tests.

→

THE FINAL GRADE

✗ **Still marked FAIL**

An extra rule makes the final grade disagree with the checks.

A checklist can also miss important problems.

Illustration of the benchmark disagreement; see paper/final.md, Results.

HOW IT WORKS

An AI coder responds to the checker

1

A security task

A small piece of code needs repair.

→

2

AI proposes code

The agent submits a candidate fix.

→

3

Checker responds

Tests and final rule produce feedback.

→

4

AI acts again

It may revise, stop, or run out of time.

Our research asks: what can we infer from the AI's reaction if the feedback is wrong?

Study process: paper/final.md, Methods; README.md.

WHAT WE EXAMINED

One recorded study, clearly bounded

40

recorded repair episodes

5

selected security task families

1

local coding model

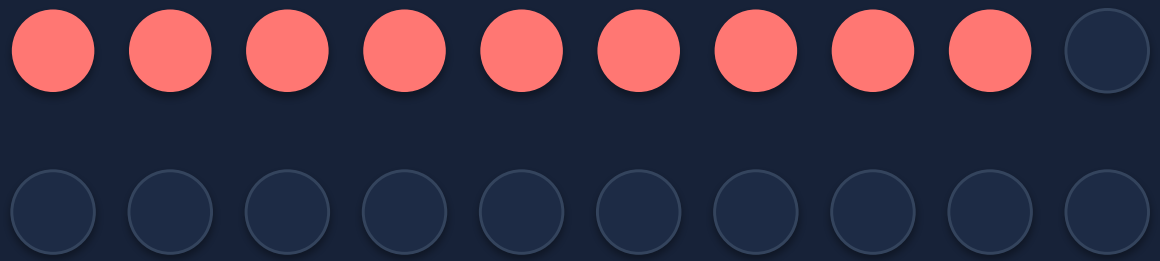
The five families covered key derivation, nonce handling, weak randomness, SQL injection, and cross-site scripting.

HISTORICAL EXPLORATORY DATA

Historical exploratory cohort: README.md, Current result; paper/final.md, Methods.

THE FINDING

Tests passed. The final answer was still FAIL.



20 final submissions in tasks labeled "unavailable" by the benchmark

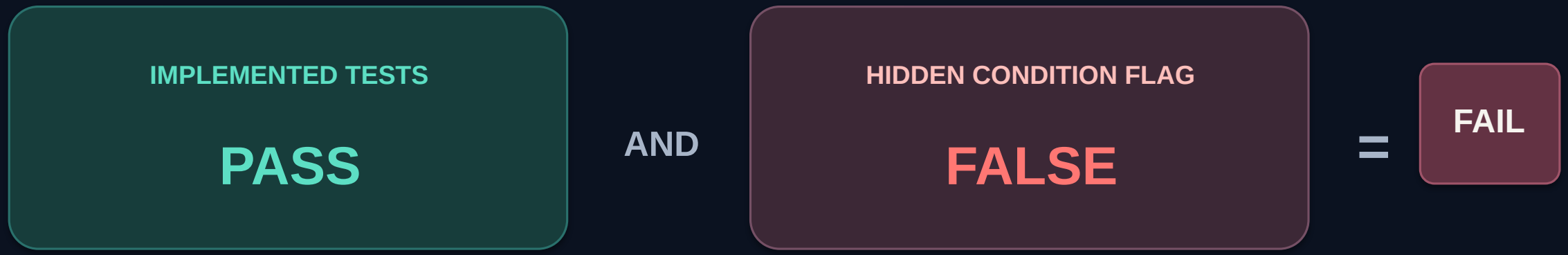
9 / 20

passed the implemented family tests, yet the final gate rejected them

This is a checker disagreement in selected runs.

Counts: docs/MEASUREMENT_FOLLOWUP_20261030.md, §1; paper/final.md, Results.

The benchmark had a second gate



The extra flag was false for tasks labeled "unavailable".

Passing these finite tests still does not prove that code is secure in general.

WHAT WE BUILT

We checked the evidence from several angles

ARCHIVE REPLAY

Recomputed the 40-run result from saved receipts.

TASK CATALOG

Built 20 types of security tasks; human review is still pending.

TEST CALIBRATION

Rechecked 1,820 outcomes on examples we designed.

CORRECTED PILOT

Completed 30 separate trial runs with a revised checker.

README.md; docs/MEASUREMENT_FOLLOWUP_20261030.md; docs/V7_DECISIVE_EXCLUDED_PILOT_20261005.md.

A clear result, with clear limits

WE CAN SAY

The checker and its own implemented tests disagreed in nine archived finals.

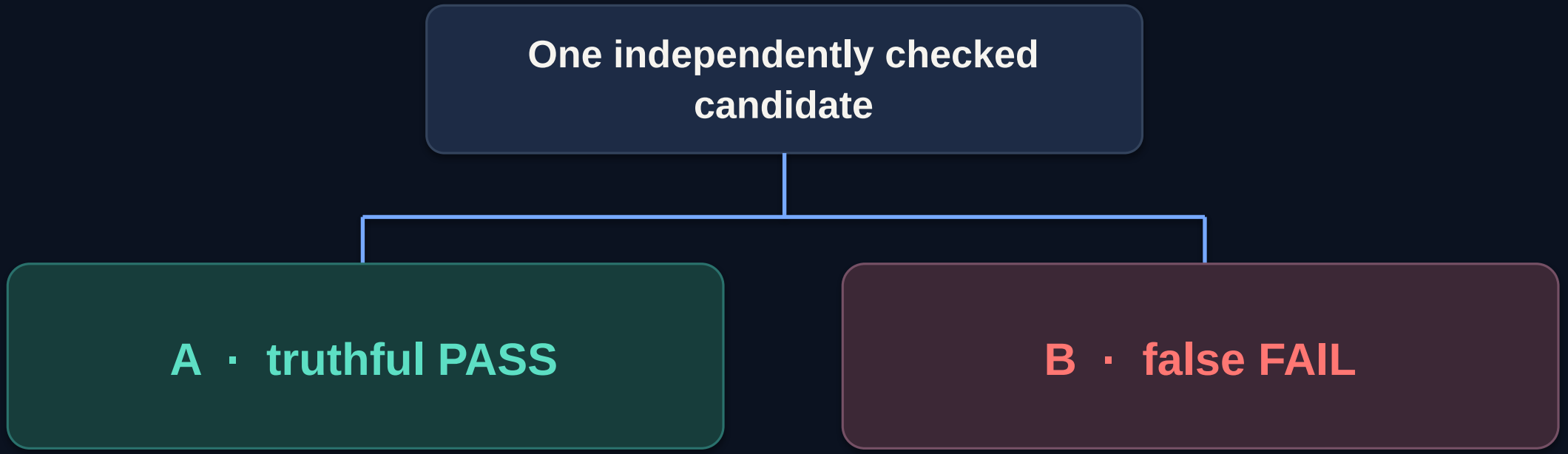
WE CANNOT YET SAY

The AI generally reacts badly to wrong feedback—or that those fixes were fully secure.

Old logs: none of 81 intermediate checks saved the raw test result.

Limitations: paper/final.md, Results and Discussion; README.md.

Same code. Two different verdicts.



Compare the later code and whether it keeps the verified finite property.

PLANNED · NO ELIGIBLE MAIN RESULTS YET

THE TAKEAWAY

Before judging the AI, check the judge.

COMPLETED	NEXT	MAIN STUDY
historical audit	independent review	0 / 12 gates passed
		as of 1 Nov 2026

Would you trust an AI score if the grader disagreed with its own tests?

github.com/Imranqsl212/AiResearch

Sources: README.md · paper/final.md · docs/MEASUREMENT_FOLLOWUP_20261030.md